COMPOSITIONS VANISHING ON COHOMOLOGY FACTOR THROUGH TRUNCATIONS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4354-4378, following the statement at 4328-4352.

Editorial note. The general case repeats the n = 2 pattern verbatim ('exactly the same manner'), matching the established induction-repetition note practice; the dual statement is the mirror of the executed argument.

Allowed standard prerequisites (editorial).

- Truncation triangle. remark-truncation-distinguished-triangle (same chapter, cited at ~4362). Provides tau_{<=-1}K -> K -> H^0(K)[0] used in the step.
- Representable functor is homological. lemma-representable-homological (same chapter, cited at ~4369). Converts the vanishing composition into the desired factorization.

Source reference. Original lemma 0149: sources/stacks-derived/pinned-original-source.tex, source lines 603–603. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 08J5: sources/stacks-derived/pinned-original-source.tex, source lines 4293–4293. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let*

$$K_0^\bullet \to K_1^\bullet \to \ldots \to K_n^\bullet$$

be maps of complexes such that

(1) $H^i(K_0^\bullet) = 0$ *for* $i > 0$,
(2) $H^{-j}(K_j^\bullet) \to H^{-j}(K_{j+1}^\bullet)$ *is zero.*

Then the composition $K_0^\bullet \to K_n^\bullet$ factors through $\tau_{\leq -n}K_n^\bullet \to K_n^\bullet$ in $D(\mathcal{A})$. Dually, given maps of complexes

$$K_n^\bullet \to K_{n-1}^\bullet \to \ldots \to K_0^\bullet$$

such that

(1) $H^i(K_0^\bullet) = 0$ *for* $i < 0$,
(2) $H^j(K_{j+1}^\bullet) \to H^j(K_j^\bullet)$ *is zero,*

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then the composition $K_n^\bullet \to K_0^\bullet$ factors through $K_n^\bullet \to \tau_{\geq n} K_n^\bullet$ in $D(\mathcal{A})$.

Proof. The case $n = 1$. Since $\tau_{\leq 0} K_0^\bullet = K_0^\bullet$ in $D(\mathcal{A})$ we can replace $K_0^\bullet$ by $\tau_{\leq 0} K_0^\bullet$ and $K_1^\bullet$ by $\tau_{\leq 0} K_1^\bullet$. Consider the distinguished triangle

$$\tau_{\leq -1} K_1^\bullet \to K_1^\bullet \to H^0(K_1^\bullet)[0] \to (\tau_{\leq -1} K_1^\bullet)[1]$$

(Remark 08J5). The composition $K_0^\bullet \to K_1^\bullet \to H^0(K_1^\bullet)[0]$ is zero as it is equal to $K_0^\bullet \to H^0(K_0^\bullet)[0] \to H^0(K_1^\bullet)[0]$ which is zero by assumption. The fact that $\mathrm{Hom}_{D(\mathcal{A})}(K_0^\bullet, -)$ is a homological functor (Lemma 0149), allows us to find the desired factorization. For $n = 2$ we get a factorization $K_0^\bullet \to \tau_{\leq -1} K_1^\bullet$ by the case $n = 1$ and we can apply the case $n = 1$ to the map of complexes $\tau_{\leq -1} K_1^\bullet \to \tau_{\leq -1} K_2^\bullet$ to get a factorization $\tau_{\leq -1} K_1^\bullet \to \tau_{\leq -2} K_2^\bullet$. The general case is proved in exactly the same manner. $\square$